Score	Story Segment
–	**Prompt:** In the middle of the night, a piece of paper is slipped under your door. It says, "DUCK!"
13.82	Confused, you grab the paper and open the door . Looking down the apartment complex corridor you see more and more doors open with perplexed heads poking out one by
3.00	one . No trace is left of whoever put these papers there, they just seem to have appeared all at once . "Duck?" Asks that one neighbour that you have rarely ever spoken to . "Yeah I got it too," you reply, "What do you think it means?" "Dunno" he
13.82	replies as you both turn towards the now gathering crowd . You head towards the crowd, which is incredibly loud with chatter now . "What's the big deal?" "Why should we even care?" Everyone's thinking the same thing yet everyone seems inexplicably interested . Amidst the crowd you notice that
13.82	the last door is still closed . You point in the direction and immediately the crowd goes silent . Everyone begins to walk over to the closed door when the knob turns . It's dark inside . All you can see is a dark figure inside.As it creeps forward you
	notice an eerie grin . A woman creeps out of the dark room . She looks up, straight into your eyes . Your heart beats faster and faster as she holds up her paper . It reads: "Goose."

Table 2. An example of a human-written story with both high-scored and low-scored segments. Scores were computed using GPT-4. The story contains a twist ending which is scored high. The high-scored segments correspond to the key plot elements and turning points, while the low-scored one is just an extension of the previous plot.

the echoes produced given longer prefixes (e.g. the echo highlighted in Figure 1(a)).

The described procedure for computing *Sui Generis* scores involves neither human judgment nor a large database for matching the generated text. All that is needed are the LLM's own generations and its own judgment of whether a story segment or its equivalent is contained in an alternative story. Table 2 shows a story with its segment-level *Sui Generis* scores. We show in the experiments that, compared to human-written stories, segments of LLM-generated stories have lower scores; in other words, these plot segments tend to be generated repetitively by the LLM, though possibly in a different order or in different parts of the story.

A qualitative user study involving expert writers (7) indicated that LLM-generated stories may differ from human-written ones in narrative pacing: In human-written stories, a plot twist is usually carefully foreshadowed and developed in order to maintain tension, which is in line with the tendency for uniform information density in language production (20). By contrast, we observe that LLM-generated stories often rapidly accelerate over time without fully resolving the plot.

Armed with the *Sui Generis* score, we can verify this observation quantitatively. Specifically, given the story segments $s_{1\ldots n}$ through time 1 to n, we take the *Sui Generis* scores $\text{SG}_{1\ldots n}$ and compute the "drop ratio" between the scores of consecutive segments clipped by a threshold θ:

$$\text{drop}_i = \max\left(\frac{\text{SG}_i - \text{SG}_{i+1}}{\text{SG}_i} - \theta, 0\right) \quad [4]$$

Intuitively, a high drop ratio indicates that the *Sui Generis* score decreases immediately after the peak at position i, without progressively unraveling the surprising plot.

4. Experimental Setup

A. Datasets. We test our *Sui Generis* score on 100 stories from two story datasets: 1) the WritingPrompts dataset (21), which contains story prompts and the corresponding human-written stories posted on an online forum[‖] by the year of 2018, and 2) plot summaries of TV episodes in the year of 2023 crawled from Wikipedia.

For WritingPrompts, we randomly sample 50 stories and directly use the story prompts by humans for LLM story generation.[**] To account for the fact that the paragraph segmentation of stories generated by different human authors and LLMs can vary significantly, thus affecting the segment-level scores, we break both human- and LLM-generated stories into segments of 50 words.[††] Examples of such segments are shown in Figure 1 within square brackets [...]. This results in a total of $1,887$ segments in the original and LLM-generated stories for WritingPrompts.

On 50 randomly sampled Wiki plot summaries, we use the first two sentences from each plot summary as the prompt for LLM story generation. We segment both human and LLM stories by a fixed length of 30 words, which is the average length of a sentence in Wiki plot summaries.[‡‡] We test on a total of $1,862$ segments from the original and LLM-generated stories.

B. *Sui Generis* Scoring Setup. We sample $K = 20$ alternative continuations at each position in the story. We set $\lambda = 0.9$ when computing the weighted average of the negative log of echoes for the *Sui Generis* score in Eq.3. To compute the drop ratio, we set the threshold $\theta = 0.5$.

C. Large Language Model Setup. We use GPT-4 (22) and LLaMA-3-70B-Instruct (23) (more details in the Appendix) with $top_p = 1.0$, sampling temperature $\tau = 1.0$ for story generation.[§§] When computing the *Sui Generis* score, we use the same story generation model (with $top_p = 0.5$ and $\tau = 1.0$) to generate alternative continuations C_j, and GPT-4 with $top_p = 0.5$ and $\tau = 0$ to estimate the function $a(s, C)$ (plot entailment), i.e. if similar plot in the original story segment appears in the alternative continuations.

5. Results

***Sui Generis* Score: Human vs. LLMs** We first compare the average *Sui Generis* scores of story segments produced by humans versus LLMs (including GPT-4 and LLaMA-3). As shown in Figure 3, for both datasets, human-written stories yield significantly higher *Sui Generis* scores than LLM-generated ones. We also show the average Echo score matrix $p_{i,j}$ on Wiki data in Figure 4 (see the other heatmap figures in the Appendix). For LLM-generated stories,

‖https://www.reddit.com/r/WritingPrompts/

**The prompt templates used for LLM generation are listed in Appendix. We also tried using GPT-4 to generate 20 different instruction prompts and evaluated the stories generated using these prompts. Results show very small variance (with a standard deviation of 1.1) in *Sui Generis* scores among stories generated with different instruction prompts.

††We chose word-level segmentation so that information is distributed evenly across segments. And the average number of words in each paragraph of WritingPrompts stories is around 50.

‡‡Because the Wiki plot summaries are more condensed than WritingPrompts stories, and usually each sentence can cover very rich plot.

§§As (24) find that the influence of temperature on the creativity of LLMs is weak.

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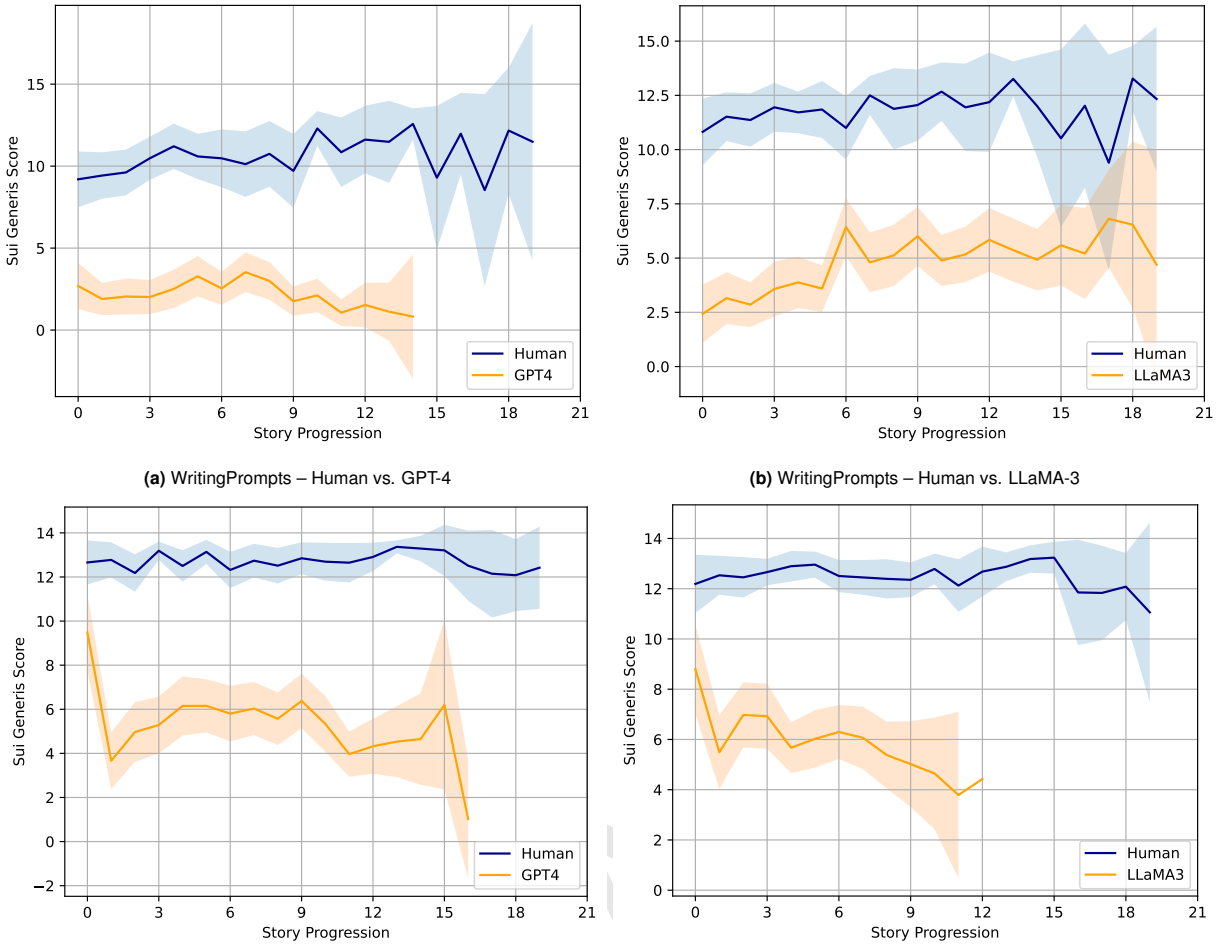


(a) WritingPrompts – Human vs. GPT-4

(b) WritingPrompts – Human vs. LLaMA-3

(c) Wiki – Human vs. GPT-4

(d) Wiki – Human vs. LLaMA-3

Fig. 3. Average *Sui Generis* scores of story segments generated by humans versus GPT-4 (in (a) and (c)) or LLaMA-3 (in (b) and (d)) at varying segment positions (*x-axis*). The shaded area represents the confidence interval with confidence level $\gamma = 0.95$.

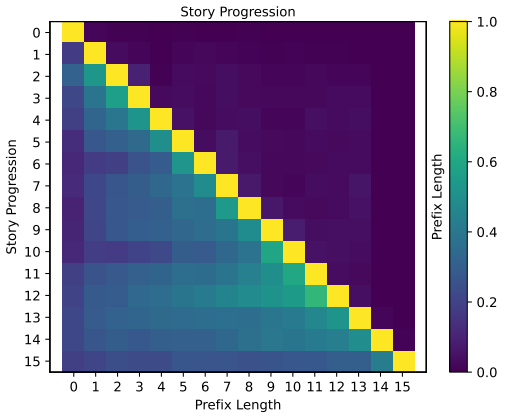
the Echo rate increases as more of the previous plot is revealed (i.e., as the prefix length increases), while for human-written stories, the LLM can only occasionally predict the beginning section of the story. These results indicate that LLM-generated stories are largely composed of idiosyncratic plot elements that are echoed across multiple LLM generations, while human-written stories often contain plot elements that land far in the tail – or not at all – of LLMs' output distribution.[¶¶]

***Sui Generis* Score: Cross-Model Scoring** Next, we test if there are common echoes across different LLMs. To this end, we compute the cross-model *Sui Generis* scores – scores of LLaMA-3 generated stories given the alternative continuations generated by GPT-4. Results on 20 Wiki and 20 WritingPrompts stories show that the cross-model *Sui Generis* scores are slightly higher than the original scores, but the differences are small (0.6 on Wiki stories and 1.6 on WritingPrompts) compared to the gap between human and LLaMA-3 story scores (> 6.0). As shown in Figure 5, the cross-model Echo scores distribute similarly to the original Echo scores. These results indicate that the plot elements generated by LLaMA-3 are echoed not only in its own generations, but also in the alternative continuations generated by GPT-4. In other words, the plots generated by LLMs are echoed frequently across different models, while human-generated plots are rarely echoed by any of these LLMs.

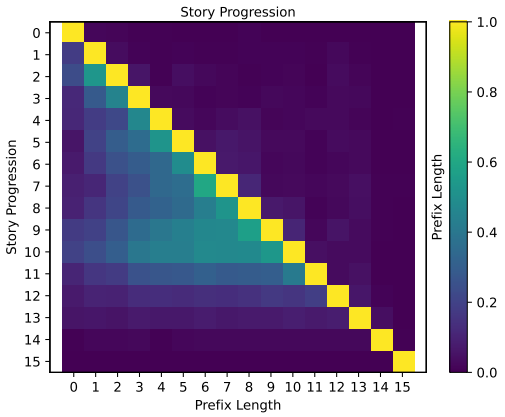
Drop Ratio: Human vs. LLMs As shown in Table 3, LLM-generated stories lead to 7–9 higher drop ratio than human-written ones. This indicates that, in LLM-generated stories, there are more cases where the *Sui Generis* score drops immediately after a peak, while in human-written stories, the score often peaks and then stays high. We further show that this is related to the overly fast pacing in LLM-generated stories in Section 6.

***Sui Generis* scores correlate with human judgment of surprise level** We invite four human judges, fluent in English, to evaluate the surprise level of story segments in 9 randomly selected stories (159 segments) generated by LLaMA-3 based on the story prompts from WritingPrompts. The study was approved by Microsoft Research's Institutional Review Board (case number 11035). Each human annotator in the study had signed a consent form electronically, informing them of the task and potential risks. Specifically, we ask each human annotator to read through each story segment by segment and annotate how surprising each segment is from level 1–3 (1-most-anticipated, 2-neutral, or 3-most-surprising) based on the story so far. To enhance agreement among annotators, we give them 5 examples (excluded from the formal annotation task) and ask them

[¶¶] We further discuss the memorization effect in Section 7.



(a) Wiki – Human vs. GPT-4



(b) Wiki – Human vs. LLaMA-3

Fig. 4. Heatmap of the average Echo scores $p_{i,j}$ for the i-th story segment given story prefix S_{-j} on 50 Wiki stories. The lower left triangle shows $p_{i,j}$ on LLM (GPT-4 or LLaMA-3) generated stories, while the upper right triangle shows the transpose matrix of $p_{i,j}$ on human stories.

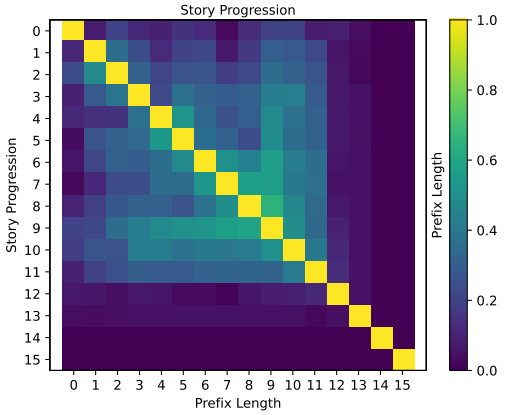


Fig. 5. Heatmap of the average Echo scores $p_{i,j}$ for the i-th story segment given story prefix S_{-j} on 20 Wiki stories by LLaMA-3. The lower left triangle shows the original Echo scores $p_{i,j}$, while the upper right triangle shows the transpose matrix of cross-model Echo scores $p_{i,j}$ (using GPT-4 to generate alternative continuations).

to discuss their annotations to reach an agreement prior the formal annotation task. For the formal annotation task, the inter-annotator agreement based on Krippendorff's alpha is moderate (0.68). Results show that segment-level *Sui Generis* scores correlate moderately with the average human judgment of surprise level: the Spearman's rank

	Human	LLM
WP: Human vs GPT-4	3.7	11.3
WP: Human vs LLaMA-3	1.7	9.0
Wiki: Human vs GPT-4	0.4	9.1
Wiki: Human vs LLaMA-3	0.5	7.8

Table 3. Comparing the drop ratio (in percentage) of stories generated by humans versus GPT-4 or LLaMA-3 on WritingPrompts (*WP*) and Wiki data. LLM-generated stories obtain consistently higher drop ratio than human-written ones.

correlation between *Sui Generis* scores and the average human rating is significant[***] and the magnitude is 0.55. The correlation between *Sui Generis* scores and the average human rating is even stronger than that between each individual human rating and the average among the other three annotators (with Spearman's correlation between 0.38 and 0.49). The relationship between human ratings and *Sui Generis* scores is visualized in Figure 6, where average human ratings (y axis) are shown for segments whose *Sui Generis* scores fall within a bin within a 1.4 interval around the x axis value. While humans rate segments with a wide range of mid-level *Sui Generis* scores as mildly surprising, they agree with *Sui Generis* on segments with highest and lowest scores.

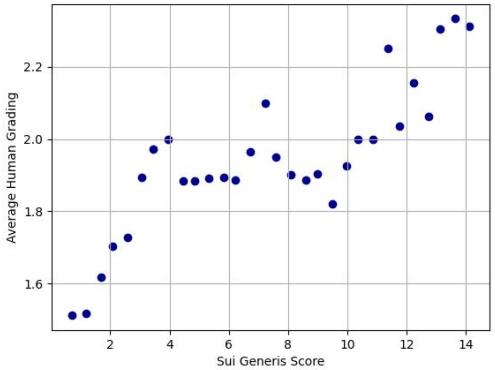


Fig. 6. Average human grading of surprisal versus the *Sui Generis* scores on story segments generated by LLaMA-3 based on the prompts from WritingPrompts. The data points are binned by an interval of around 1.4 based on the *Sui Generis* scores.

The story prompt itself affects *Sui Generis* scores for both human and LLM stories We hypothesize that the story prompt will have similar impact on the *Sui Generis* scores of stories generated by humans and LLMs, as certain story ideas in the prompt (e.g. in Figure 1(c)) simply lead themselves to more interesting and diverse continuations. To test the hypothesis, we compute the Spearman's correlation between the *Sui Generis* scores of pairs of human-written and LLM-generated stories under the same prompt. The correlation between the scores of human- and GPT-4-generated stories is 0.48 with p-value < 0.01, which indicates a moderate correlation. The correlation between human- and LLaMA-3-generated stories is 0.34 with p-value < 0.01, which indicates a weak but significant correlation. This suggests that, although human-written stories are scored significantly higher than LLM-generated ones, story prompts that lead to relatively low-scored stories by LLMs also lead to low-scored stories by humans. Thus, one can boost the *Sui Generis* score of human and LLM-generated stories by improving the story prompts; prompting is itself a creative endeavor.

[***] We use the p-value threshold $p = 0.05$ for all the significance test in this paper unless noted otherwise.